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Corrosion studies of LiH thin films

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HIGHLIGHTS

- Thin films of high purity LiH, LiOH and Li_2O have been produced.
- O1s XPS shows separate peaks for LiOH and Li_2O.
- LiH exposed to low H_2O doses shows the presence of both LiOH and Li_2O.
- With higher doses, chemisorbed H_2O is observed.
- Decomposition kinetics of thin films are similar to previous bulk studies.

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ABSTRACT

Thin films of LiH and its corrosion products were studied using temperature programmed decomposition (TPD), x-ray photoelectron spectroscopy (XPS) and Auger electron spectroscopy (AES). Thin films were grown on Ni(100) in an ultra high vacuum system using an electron beam evaporator. Characteristic Li KLL AES peaks were identified for Li, LiH, Li_2O and LiOH which facilitated identification of thin film composition. XPS of the O 1s region revealed three distinct chemical shifts which were attributed to Li_2O, LiOH and chemisorbed H_2O. We show that exposing LiH to very low H_2O partial pressures results in formation of LiOH/Li_2O domains on LiH. We also show that these XPS peaks can be linked to reaction mechanisms in the TPD profiles. TPD traces have been explicitly modelled to determine the activation energies of the reactions and compare favourably with previous measurements on bulk LiH samples.
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1. Introduction

Lithium hydride (LiH) has several attractive properties that have sustained interest in the material for well over a century. From a theoretical point of view, LiH is the lightest molecular solid and its simple electronic structure means that the material offers an excellent test of first principle calculations [1–6]. From a practical point of view, LiH finds application as a hydrogen storage material [7], a strong reducing agent [8] and in the nuclear industry as an efficient neutron moderator [9–11].

The labile reaction between LiH and H_2O (equation (1a)) is of interest in the aforementioned practical applications due to its deleterious effects on the hydrogen density, increased oxygen content, and an increase in the presence of H_2 gas when in a hermetic environment. Consequently, LiH hydrolysis has produced a wealth of literature which will now be summarised. The current understanding of LiH hydrolysis involves the formation of a trilayer system of LiOH/Li_2O/LiH (bulk), which is formed by the reactions shown in equation (1) [8,12–16].

$$2LiH + H_2O \rightarrow Li_2O + 2H_2 \quad (1a)$$

$$LiH + H_2O \rightarrow LiOH + H_2 \quad (1b)$$

$$Li_2O + H_2O \rightarrow 2LiOH \quad (1c)$$

Experimental evidence tends to favour the formation of a Li_2O layer on LiH for low H_2O exposures (up to 1 ML (monolayer)) [12,15,17]. Theory also points toward equation (1a) as the enthalpy of formation of this process is more favourable compared to equation (1b) [8,15]. The subsequent thermal decomposition of the trilayer LiOH/Li_2O/LiH (bulk) system causes the release of H_2O and H_2. There are two proposed decomposition routes leading to the formation of a surface layer of Li_2O only. These processes are given

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in equation (2) [8,14,18,19].

$$2LiOH \rightarrow Li_2O + H_2O \quad (2a)$$

$$LiH + LiOH \rightarrow Li_2O + H_2 \quad (2b)$$

The key aspect of the current investigation with respect to previous work is that LiH specimens here are fabricated in a clean environment, within an ultra high vacuum (UHV) chamber. To the authors knowledge, all previous hydrolysis studies have involved powder or bulk specimens [14,16,18–23], the latter being produced from powder compaction. Given the high reactivity of LiH to moisture, it is inevitable that impurities will be incorporated within the bulk material (for LiH this is primarily O), even in dry Ar glovebox conditions [24]. The work presented here therefore has the advantage of sample purity and associated identification of chemical species over previous studies.

In order to investigate the fundamental chemistry of LiH hydrolysis we have produced pure LiH films and exposed them to known quantities of moisture. The use of thin films and model systems has been very successful in developing a mechanistic description of many surface processes, for example: heterogeneous catalysis [25–29], thin film photovoltaics [30] and in the growth of thin film semiconductors [31]. The work described here is a study of the fundamental interactions of H_2O with pure LiH thin films and the nature of the solid state corrosion processes. This has been achieved with several surface sensitive probes (x-ray photoelectron spectroscopy (XPS), Auger electron spectroscopy (AES) and temperature programmed decomposition (TPD)) and provides interesting new insight into the hydrolysis phenomena.

2. Experimental

This work was carried out using an UHV system with a base pressure of 10^{-8} Pa which has been described in more detail elsewhere [32]. The system is equipped with XPS and AES for spectroscopic analysis, low energy electron diffraction (LEED) for surface crystallography and TPD for thermal analysis. Also present is an electron beam deposition source for metallic thin film growth and leak valves for accurate gas dosing.

The hydriding of Li films described here is based on the method developed by Engbæk et al. [33] and is outlined below. A Ni(100) single crystal (Goodfellow™ Cambridge Ltd, UK) was chosen as the substrate primarily because it is immiscible with Li and Li does not dissolve into the bulk upon heating [34]. The Ni(100) crystal was mounted on a Mo sample holder that could be resistively heated to 1100 K and cooled with liquid N_2 to 180 K. In preparation to deposit Li, Ni(100) was cleaned using 3 kV Ar ions with a sample current of 20 μA for 10 min and then annealed to 900 K. This cleaning cycle was repeated until no O 1s, C 1s or Li *KLL* peaks were detectable and a sharp (1 × 1) LEED pattern observed.

Li was deposited on the Ni(100) using the electron beam source, and the flux recorded with a picoammeter. The flux was calibrated to a thickness in ML using known LEED patterns. A (2 × 2) overlayer structure equating to 0.25 ML and a (5 × 5) overlayer structure equating to 0.55 ML was used to provide a two point calibration [35]. Furthermore the attenuation of the Ni *MNN* Auger line at 846.7 eV was used to estimate Li film thickness. The thickness was calculated using $t = \lambda \cos(\theta)\log(I_0/I)$, where λ is the electron inelastic mean free path length in Li ($\lambda \approx 3.5$ nm at 846.7 eV [36]), θ is the take-off angle, I_0 is the clean Ni signal intensity and I is the Ni intensity with a Li film deposited on top [37]. The Li film was then exposed to H_2 at 200 Pa to an exposure of >1000 L (1 L = 10^{-6} torr s = 1.33 × 10^{-4} Pa s) at room temperature to create LiH via the direct combination of the elements (*i.e.* $2Li + H_2 \rightarrow 2LiH$). H_2 was produced from the electrolysis of demineralised H_2O. Hydrolysis studies were performed by allowing H_2O vapour through a leak valve, which in turn originated from a heated flask of freeze-pump-thawed demineralised H_2O.

XPS and AES have been carried out on all of these thin film compounds in order to determine the surface composition. XPS was performed using an Al K_α ($h\nu$ = 1486.7 eV) source and a concentric hemispherical analyser (CHA) to determine the kinetic energy of the emitted electron. A 2 point calibration of the XPS energy scale was done with the Ni $2p_{3/2}$ and the C 1s lines at 852.6 and 285.0 eV, respectively. AES used a 12 kV electron source operating with a sample current of approximately 10 nA and the same CHA as used for XPS. This surface chemical analysis was followed by TPD from ambient to 1000 K at a variety of heating rates to investigate the mechanisms and kinetic parameters of LiH hydrolysis and subsequent thermal decomposition. The heating rates used for TPD were 0.2, 0.5, 1 and 2 K s^{-1}.

3. LiH thin films

After synthesis, the Li thin film was immediately characterised with AES and XPS. Previous work shows that the AES spectrum of Li exhibits a single transition, the Li *KLL* line at 53 eV which is shifted to between 38 and 41 eV for oxides [20,38–42]. This is sufficiently separated from the Ni *MNN* transition at 60 eV to allow it to be clearly observed. The black line in Fig. 1 shows the Li *KLL* AES spectrum of a typical Li film; which shows a large Li signal at 53 eV in comparison to the attenuated Ni *MNN* signal at 60 eV [33,38]. For LEED patterns of Li greater than 0.55 ML, a gradual attenuation of the 5 × 5 pattern was observed, indicating three dimensional growth above this coverage.

Upon exposure of the Li film to 200 Pa H_2, a shift in the *KLL* line from 53 eV to 44.8 eV (the red line in Fig. 1) is observed with minimal contamination of the sample by O and C. In order to obtain AES spectra for Li_2O, a Li film was exposed to 10^{-4} Pa O_2 for 120 s. For LiOH, this Li_2O film was exposed to 10^{-4} Pa H_2O for 360 s. The AES spectra for Li_2O and LiOH are shown in Fig. 1. The AES spectrum for Li_2O shows a strong feature at 42 eV close to what has been previously reported. To the authors knowledge no high resolution Li *KLL* data for LiOH has been reported. AES spectra for LiH, Li_2O and LiOH were used throughout this study to corroborate XPS and TPD experiments. No structural information could be obtained from LEED patterns of LiH films. This indicates that the films generated were either amorphous, three dimensional or a combination of both.

From the two point LEED calibration of the evaporation source and the analysis of the Ni Auger attenuation, the thickness of Li films examined here were estimated to be 10–12 ML. XPS was unable to resolve the Li 1s emission due to its relatively low photoionisation cross-section, and close proximity to the Ni 3p emission. However a survey spectrum revealed that the C and O contamination was minimal at ~3 at%, whilst the adventitious O could be resolved into LiOH and Li_2O components at a ratio of approximately 1:1 (Fig. 2a), with a Guassian/Lorentzian product peak shape (30% Guassian) and full width half maximum (FWHM) values of 2.18 and 1.43 eV, respectively. O contamination is likely to be from residual H_2O in the H_2 supply as the pressure during Li deposition was less than 2 × 10^{-7} Pa. The observation of co-existing Li_2O and LiOH indicates that even at very low H_2O exposures, the LiOH/Li_2O/LiH trilayer corrosion model remains valid. The high reactivity of H_2O with Li_2O would suggest that H_2O mobility on a Li_2O surface is low, effectively promoting the formation of LiOH islands. In their recent paper, Guichard et al. [17] examined the hydrolysis of bulk LiH under low humidity. The authors concluded that no LiOH could be observed below a pressure of 1.3 Pa, which is

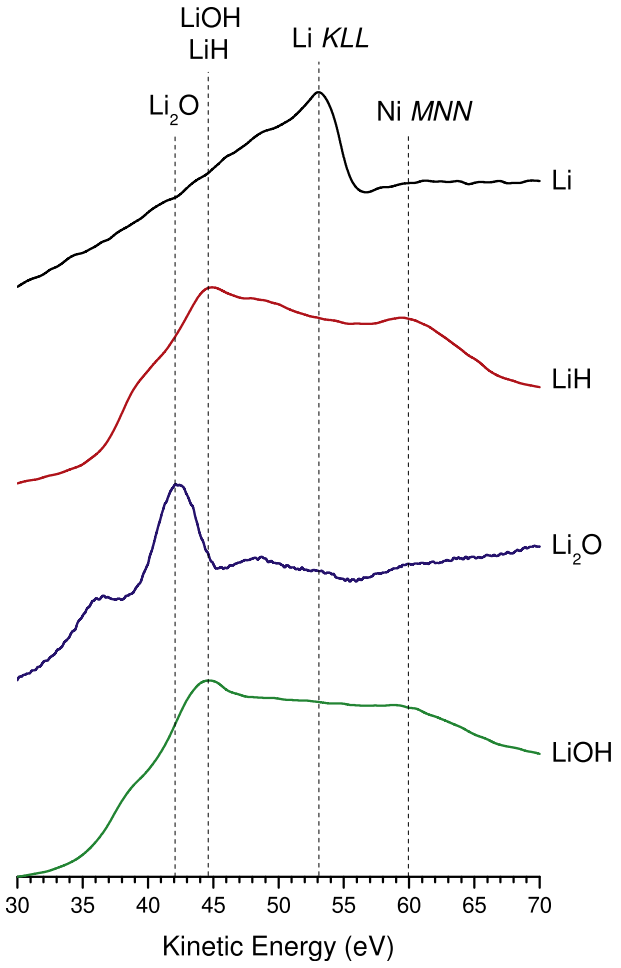


Fig. 1. AES spectra of Li (black), LiH (red), Li_2O (blue) and LiOH (green) films on Ni(100). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

several orders of magnitude higher than the H_2O pressure during experiments here, approximately 10^{-6} Pa. This apparent contradiction can be explained by recognising that Guichard et al. used probes (x-ray diffraction and infra-red spectroscopy) which are not surface sensitive and therefore would not be able resolve any trace hydroxide species. Indeed, the authors point out that the formation of LiOH is possible if its concentration were less than the detection limit of the techniques used.

Prior to LiH thin film experiments, two control experiments were performed, with the results displayed in the $m/z = 7$ trace of Fig. 3. In the first of which, Li metal was deposited onto the Ni(100) substrate and a TPD performed (magenta dashed line in Fig. 3). This resulted in a peak at 640 K, similar to that obtained by Engbæk [33]. The second control experiment involved performing a TPD on a crystalline bulk LiH specimen, formed through the compaction of powders in a die press. This TPD (dotted green line in Fig. 3) produced a single exponential feature beginning around 775 K which was ascribed to LiH sublimation and is close to the melting point of bulk LiH [43]. Sublimation was not monitored to completion due to the concern that excess Li may damage the mass spectrometer.

TPD was then performed on the LiH thin film, the result of which is displayed in Fig. 3. Two peaks are evident from the TPD profile of H_2. One of which is associated with a Li desorption peak at 825 K, which is close to the sublimation temperature of bulk LiH indicating a high degree of crystallinity. The lack of any features at 640 K in the LiH TPD indicates that there is no free Li metal and that the film has fully hydrided. As a result of these observations we conclude that we have successfully synthesised a thin film of high purity LiH, in agreement with the assertion that the single Li *KLL* emission at 44.8 eV originates from LiH. The other H_2 peak is a broad feature centred at 425 K which is not associated with water desorption, as evidenced by the lack of any peaks in the TPD trace of H_2O. This result, combined with the O 1s XPS, indicates that very thin overlayers (less than 0.2 ML) of LiOH on LiH decompose to Li_2O to produce H_2 without H_2O. There are two possible models that explain the lack of any evolved H_2O. Firstly, the decomposition may be entirely through the solid state reaction, equation (2b). This assertion fits well with the fact that equation (2b) has been proposed to be mechanistically a solid state reaction, in which species must diffuse to the reaction interface [19,23]. An extremely thin layer will drastically reduce the distance over which chemically active species must travel before reacting and therefore would be favoured in thin layers. Also the temperature of the LiOH conversion to Li_2O is in agreement with other studies on bulk LiH compacts, indicating similar decomposition kinetics [19]. Alternatively, there may be some H_2O evolved from LiOH, and as the Li_2O layer is extremely thin, the H_2O will be entirely consumed by the LiH layer. Thermodynamically, the solid state reaction (equation (2b)) is more favourable than the thermal decomposition of LiOH to Li_2O and H_2O (equation (2a)). The heat of reaction of equation (2b) is -22.9 kJ mol^{-1} compared to 124 kJ mol^{-1} for equation (2a) at 298 K [8]. Given the more favourable thermodynamics, it is reasoned that the solid state reaction is the more probable reason for the lack of evolved H_2O.

Additional characterisation of the decomposition process was provided in the O 1s XPS spectrum obtained after heating the film to 675 K, shown in Fig. 2b. In comparison to the XPS spectrum prior to heating (Fig. 2a), the XPS spectrum collected after heating to 675 K reveals that partial conversion of LiOH to Li_2O has occurred, as the ratio is now approximately 1:4 where it was 1:1 previously. The residual LiOH component may be a result of incomplete conversion of LiOH to Li_2O. However, as samples were allowed to cool after TPD for about an hour, it is not possible to discount that the small LiOH component is a result of H_2O adsorption from the background gases in the chamber. The O 1s spectrum obtained after TPD is very similar to what has been observed for a Li_2O pellet exposed to 0.8 L of H_2O [44]. At this thickness of LiOH, it is probably better described as a $-$OH terminated Li_2O surface. This is an apparent contradiction to previous studies that report heating to modest temperatures (for example 473 K [21] and 550 K [14]) resulted in the full conversion of LiOH to Li_2O. Again, the techniques used in these studies lacked surface sensitivity and therefore it is likely that small amounts of LiOH were still present on the surface after heating but were not able to be observed.

The peak shape used for fitting was the same as prior to heating and the FWHM values here are 2.45 and 1.75 eV, respectively. This represents an increase of 0.28 ± 0.01 eV in the FWHM upon heating. There is also an increase in binding energy of 0.7 eV in the Li_2O component of the O 1s spectrum upon heating. A similar effect has been recently observed using Raman spectroscopy, in which extra features associated with the Li_2O vibration emerged after heating [45]. In this paper, the authors argued that the appearance of multiple Li_2O vibrations arise because there is a large lattice mismatch between LiH and Li_2O. As the Li_2O layer grows with heating, it is likely that to relieve the increasing interfacial strain, multiple domains of Li_2O of varying crystallinity are required. Indeed cracking and blistering has been previously observed upon heating, attributed to stress relieving mechanisms [19]. Although these studies used bulk specimens, an increase in interfacial stress and a subsequent decrease in Li_2O crystallinity is consistent with

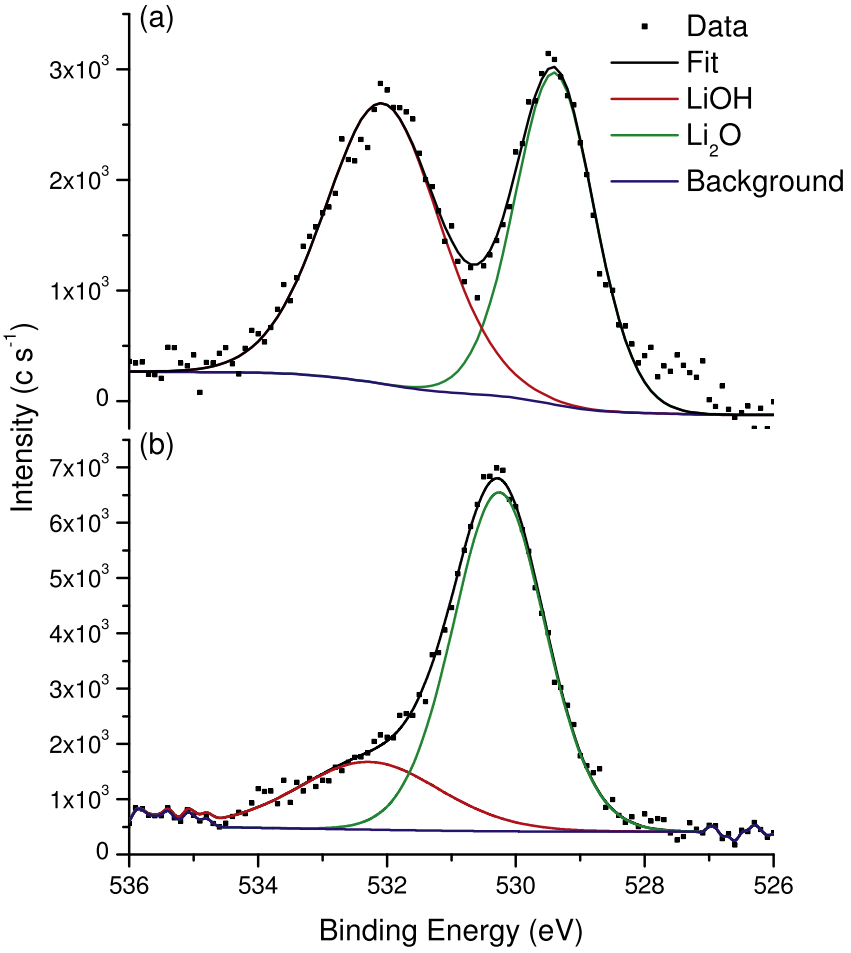


Fig. 2. XPS spectra of the O 1s region of a LiH film on Ni(100). Two peaks are present, assigned as LiOH (red) and Li_2O (green). (a) is from a LiH film as fabricated at ambient temperature and (b) is a LiH film after heating to 675 K. LiOH (red) has reduced and Li_2O (green) has increased in comparison to the film as deposited. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the O 1s spectra reported here. The O 1s peak associated with Li_2O shifts to higher binding energy, indicating that there is less oxide-like character to the Li_2O layer, in other words there are fewer O ions with a formal O^{2-} charge, which is consistent with a decrease in crystallinity. In addition, an increase in the FWHM of the Li_2O component upon heating is consistent with a decrease in the structural order of this layer. Deliberately exposing a LiH film to H_2O produces a Li_2O layer (Fig. 4b) that has an O 1s binding energy very close to that of the adventitious Li_2O at room temperature, suggesting that the ambient temperature Li_2O structure is independent of H_2O dose, within the limits investigated in this work (0–450 L at a H_2O partial pressure of 10^{-4} Pa).

4. LiH films exposed to H_2O

In order to investigate the hydrolysis reaction further, LiH samples were then intentionally exposed to increased pressures of H_2O to generate LiOH according to equation (1). LiH films were exposed to 450 and 45 L of H_2O at a pressure of 10^{-4} Pa. O 1s XPS spectra of the 450 and 45 L films are presented in Fig. 4. Fig. 4 in comparison to Fig. 2a shows that the LiOH quantity has increased as expected with exposure to moisture. In comparison with the non H_2O exposed LiH, the O 1s emission is approximately 10 times more intense in both spectra. An additional emission is observed at 535 eV in the film exposed to 450 L which is a binding energy typical for adsorbed H_2O [46,47]. No Li_2O is observed in this film which was presumed to be a result of attenuation by the $H_2O_{(ad)}$ and LiOH layers, consistent with the proposed trilayer corrosion model, in which LiOH and adsorbed H_2O form the outermost layers. With high H_2O exposures, there is evidence of some charging related binding energy shifts in the O 1s features [48]. It is not possible to categorically determine if the H_2O exists on the surface as a simple chemisorbed species or as $LiOH{\cdot}H_2O$, a compound known to exist for high partial pressures of H_2O, 520 Pa [23]. However, as the sample was dosed at 10^{-4} Pa, it appears unlikely that there would be sufficient pressure to generate the formal $LiOH{\cdot}H_2O$ compound and therefore suggests that the chemisorbed species is more likely. Indeed, a recent report has shown that exposure of a Li film at 100 K to 15 L of H_2O results in the physisorption of H_2O [49]. It is therefore plausible that LiOH could chemisorb H_2O at higher temperatures.

The 450 and 45 L films were then studied using TPD (Figs. 5 and 6 respectively) and reveal H_2O desorption features associated with the $H_2O_{(ad)}$ and LiOH O 1s peaks. All TPDs performed left a residual Li_2O layer (with a trace amount of LiOH) on the Ni substrate, as a result, the peak at 520 K was attributed to the decomposition of LiOH (equation (2a)) and desorption of H_2O at 425 K. Although XPS has been used to characterise LiOH and Li_2O corrosion layers in bulk LiH samples, to our knowledge, this is the first time that XPS has been used to study these decomposition/desorption reactions. O 1s binding energy assignments of corrosion and decomposition products are summarised in Table 1 and are similar to findings in

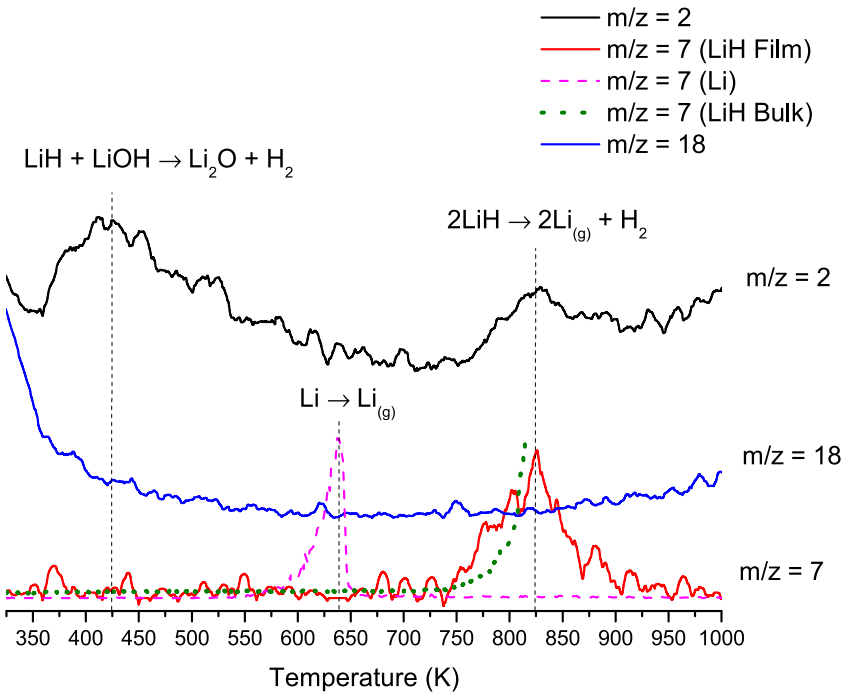


Fig. 3. TPD traces for the decomposition of thin film LiH, with thin film Li and bulk LiH decomposition given for reference. TPD traces for thin film LiH are given as solid lines (black – H_2, blue – H_2O and red – Li), a TPD trace for thin film Li is given as a dashed line (magenta – Li), and a TPD trace for bulk LiH is given as a dotted line (green – Li). A ramp rate of 0.5 K s^{-1} was used. Peaks are labelled with assigned reactions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

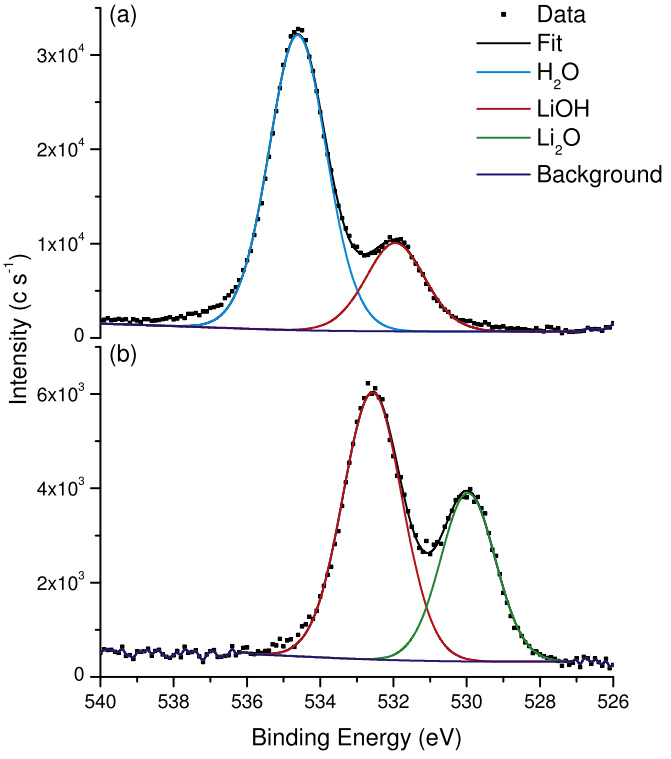


Fig. 4. XPS spectra of the O 1s region of a LiH film on Ni(100) exposed to (a) 450 L of H_2O and (b) 45 L of H_2O. Three peaks are present, assigned as adsorbed H_2O (cyan), LiOH (red) and Li_2O (green). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

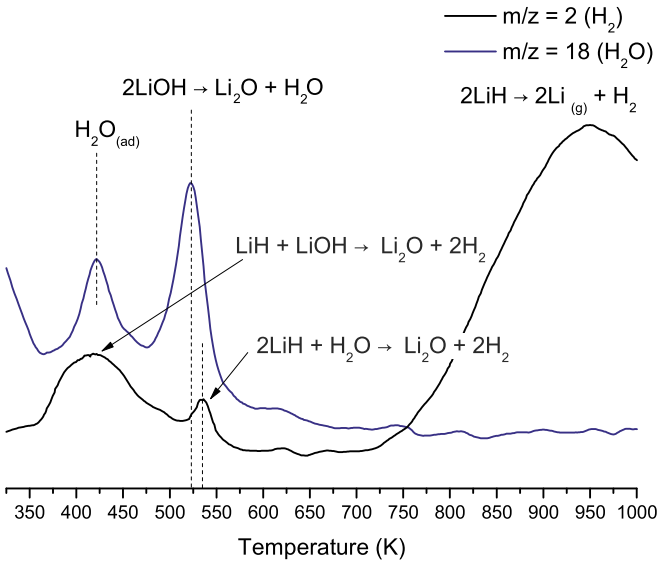


Fig. 5. TPD traces for *m/z* of 2 (H_2) in black and 18 (H_2O) in blue for a LiH film on Ni(100) exposed to 450 L of H_2O at a ramp rate of 1 K s^{-1}. Peaks are labelled with assigned reactions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

previous studies [23,44,48,50]. Also presented within this table are the fitting parameters for the peaks.

In both 45 L and 450 L exposures, LiOH decomposition at 520 K is followed by an associated evolution of H_2 from the sample at 535 K. We attribute this to be the result of reaction according to

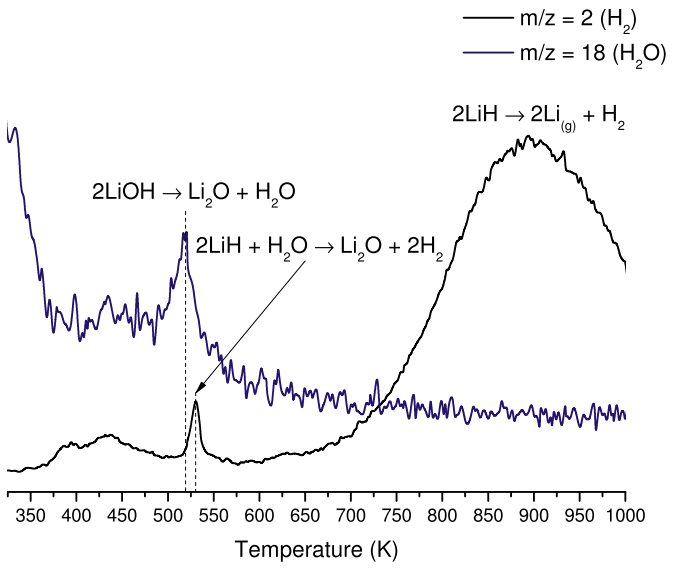


Fig. 6. TPD traces for m/z of 2 (H_2) in black and 18 (H_2O) in blue for a LiH film on Ni(100) exposed to 45 L of H_2O at a ramp rate of 1 K s^{-1}. Peaks are labelled with assigned reactions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Summary of O 1s binding energies and FWHMs for LiH corrosion and decomposition products. The peak shape used is a Guassian/Lorentzian product (30% Guassian).

Compound	Binding Energy (eV)	FWHM (eV)
Li_2O	529.4–530.3	1.68 ± 0.05
LiOH	531.9–532.6	1.79 ± 0.05
$H_2O_{(ad)}$	534.6	1.76 ± 0.05

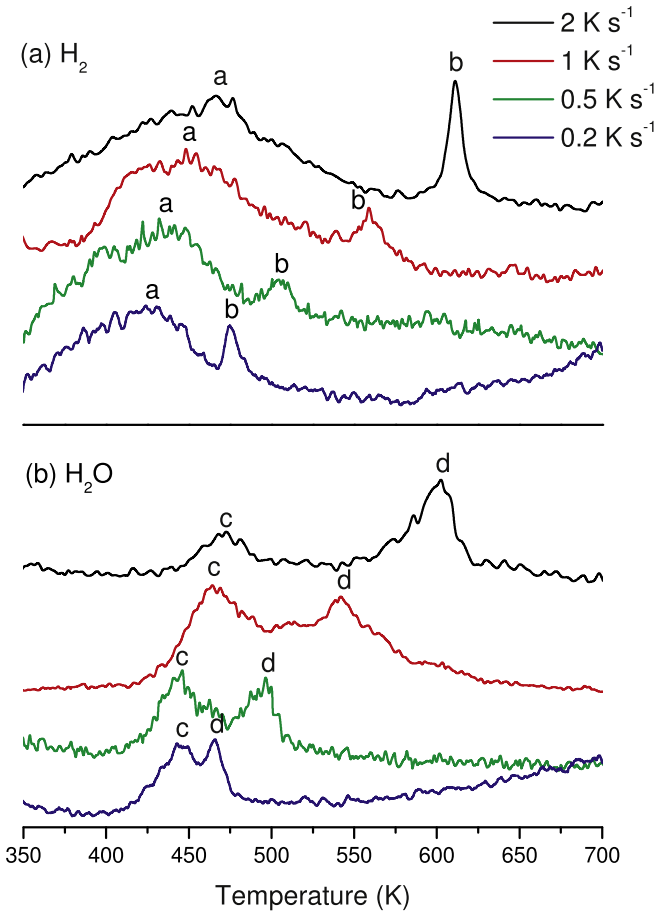


Fig. 7. H_2 and H_2O TPD traces at ramp rates of 2 K s^{-1} (black), 1 K s^{-1} (red), 0.5 K s^{-1} (green) and 0.2 K s^{-1} (blue). The peaks are labelled as 'a' - the reaction of LiH and LiOH, 'b' - the reaction of LiH and H_2O, 'c' - the desorption of $H_2O_{(ad)}$ and 'd' - the decomposition of LiOH. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

equation (1a), whereby the liberated H_2O from LiOH decomposition can react with the LiH substrate to form Li_2O and a new H_2 peak, in addition to the one produced by equation (2b). The affect of adsorbed on the H_2 evolution at 425 K is not as clearly defined as that of LiOH decomposition, as the solid state reaction between LiH and LiOH (equation (2b)) proceeds concurrently. In order to resolve these features and to elucidate kinetics, TPDs were performed at a variety of ramp rates. This causes the temperature at which reactions occur to shift as a function of the mechanism and thermodynamics of the reaction. The H_2 and H_2O traces are shown in Fig. 7 for ramp rates of 2, 1, 0.5, and 0.2 K s^{-1}. The peaks are labelled as:

a. $LiH + LiOH \rightarrow Li_2O + H_2$

b. $2LiH + H_2O \rightarrow Li_2O + 2H_2$

c. $H_2O_{(ad)} \rightarrow H_2O_{(g)}$

d. $LiOH \rightarrow Li_2O + H_2O$

Fig. 7 shows that as the decomposition of LiOH (reaction *d*) varies with heating rate, the hydrolysis of LiH (reaction *b*) also shifts temperature to be slightly higher than that of the decomposition. This provides further evidence to show that reactions *d* and *b* are linked with some of the H_2O evolved diffusing through the Li_2O layer to the LiH to cause further reaction. Another reason for varying the heating rate was to determine if peak *a*, which is quite broad, comprised of two components (solid state reaction and hydrolysis from the $H_2O_{(ad)}$). However, as both reactions shift with ramp rate by similar amounts, it was not possible to provide clarification on this. Indeed peaks *c* and *d* are sharp and therefore it is reasonable to expect a feature similar in shape to *b* to be present within the H_2 profile. This is evidently not the case and therefore it is unlikely that the $H_2O_{(ad)}$ interacts with the LiH. In summary, the LiOH layer (which is still present when the H_2O desorbs) must therefore provide a physical barrier to prevent H_2O propagation toward the LiH, causing the H_2O to desorb rather than react.

Attempts to determine the activation energy (E_a) were performed using Kissinger analysis [51], these proved to be unsuccessful for all reactions. Kissinger analysis assumes that the reaction rate can adopt the form $(1-\alpha)^n$, where α is the reacted fraction and n is the reaction order. Reactions schemes of this type are not generally valid for solid state reactions [52] and therefore may explain why the Kissinger analysis fails to adequately calculate E_a.

Explicit modelling was performed on both the H_2O and the H_2 TPD traces. The best mathematical fits were determined to be an A3 (three dimensional Avrami-Erofeev nuclei growth) mechanism for the H_2O related reactions and D3 (three dimensional spherical diffusion) for the solid state reaction [18]. D3 has also been previously identified as the best chemical mechanism for the solid state reaction within bulk LiH due to the requirement for diffusion across the Li_2O interlayer [19]. Fig. 8 shows the fits for the H_2 and H_2O traces respectively. The fit parameters are given in Table 2 where ν is a pre-exponential factor. The values determined for the E_a of the

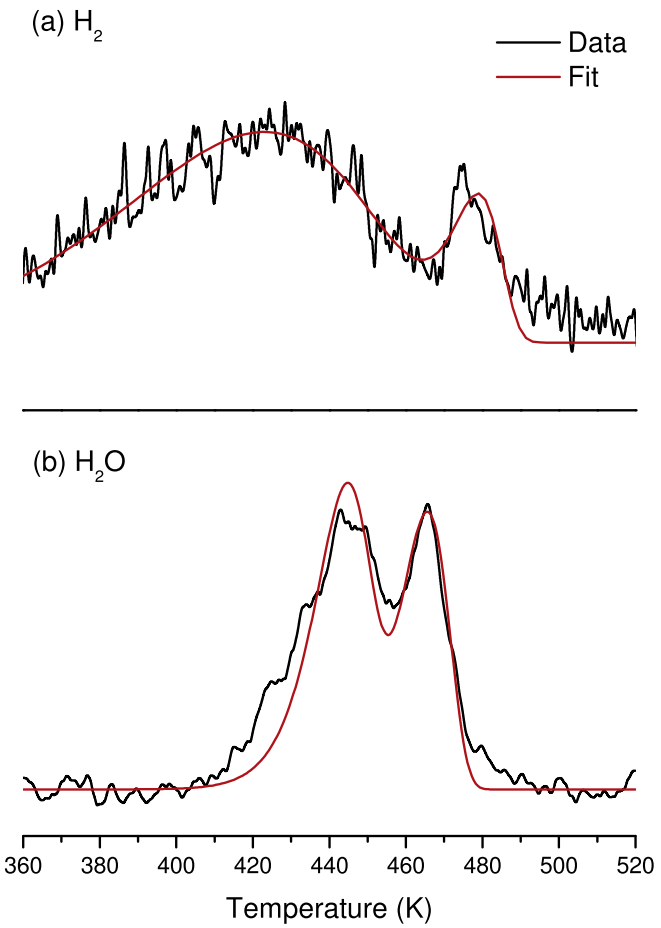


Fig. 8. Fitted H_2 and H_2O TPD traces of a LiH film exposed to H_2O at 0.2 K s^{-1}. The line in black is the data with a background correction applied and the line in red is the applied fit. The fit consists of a D3 mechanism for the solid state reaction of LiH and LiOH and an A3 mechanism for the reaction of LiH with H_2O, the desorption of $H_2O_{(ad)}$ and the decomposition of LiOH. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

decomposition of LiOH (90 kJ mol^{-1}) and the solid state reaction between LiH and LiOH (82 kJ mol^{-1}) are comparable to previous studies of the corrosion layers formed on bulk LiH which were 97 and 70 kJ mol^{-1}, respectively [18,19]. The discrepancy between these values could be a result of different mechanisms being rate limiting in thin films in comparison to the bulk; differences in thermal transport; or the use of a high purity thin film which will result in less interference from mass transport, which may have influenced the results obtained from the bulk sample in comparison to a thin film. In addition, the lack of a LEED pattern could indicate poorly ordered surface which could alter any grain boundary effects within the reaction mechanisms. As a result of the close link between the decomposition of LiOH and the reaction between LiH and H_2O, the slight increase in E_a (3.1 kJ mol^{-1}) shown here is proposed to be a result of a slight time delay in which the H_2O must diffuse across the Li_2O layer. An analogous study has been reported which is in agreement with this hypothesis. Using a H_2O molecular beam, Balooch et al. [22] showed that there is a phase lag (and therefore a time lag) between H_2 production and scattered H_2O molecules, which increases with increasing H_2O exposure. A greater phase lag for thicker LiOH and Li_2O films is consistent with a diffusion process for the transfer of H_2O to LiH. However, in the case described here, nuclei growth (A3) provided the best fit to this process, indicating that diffusion across the Li_2O region is not the

Table 2
E_a and ν determined for decomposition reactions of LiH. The error is the standard error on the mean from all ramp rates.

Reaction	E_a (kJ mol^{-1})	ν (10^6 s^{-1})
$H_2O_{(ad)} \rightarrow H_2O_{(g)}$	72.7 ± 2.8	2.3 ± 1.8
$LiOH \rightarrow Li_2O + H_2O$	90.1 ± 3.6	27 ± 2.6
$LiH + LiOH \rightarrow Li_2O + H_2$	81.6 ± 3.8	6.2 ± 2.4
$2LiH + H_2O \rightarrow Li_2O + 2H_2$	93.2 ± 2.4	52.2 ± 8.8

rate limiting process. Instead the process is governed by the decomposition of the LiOH layer.

The precise nature of the solid state reaction is a subject of debate. For example, Myers [14] and Dinh [19] have proposed mechanisms based primarily upon H_2O and H^+ transport across the Li_2O layer, respectively. Here, we reason that an alternative model is consistent for the experimental data obtained. Instead of the transport of a species across the film to react at the LiH surface, the diffusion process is equivalent to Li–O bond rearrangement in the Li_2O layer and H_2 evolution at the interfaces. Fig. 9 provides a graphical summary of the proposed model for the solid state reaction. At the LiH – Li_2O interface:

$$2LiH + Li_2O \rightarrow Li_2O + H_2 + 2Li \quad (3)$$

In essence, equation (3), describes the splitting of LiH into its constitute elements. The heat of formation of reaction 3 is 774 kJ mol^{-1} at 298 K. It is likely that this process will be mediated by the fact that the interfacial layer will be highly defective due to the large lattice mismatch and thus the heat of formation will be significantly lower. In equation (3), the Li atoms are actually switching bonds from H to O at the LiH/Li_2O interface, causing nearest neighbour shifting of Li–O bonds throughout the Li_2O layer. This will leave an excess of Li atoms at the Li_2O/LiOH interface, which converts LiOH to Li_2O. The process at the Li_2O/LiOH interface is therefore:

$$2LiOH + 2Li \rightarrow 2Li_2O + H_2 \quad (4)$$

This process has a heat of formation of −227 kJ mol^{-1} at 298 K.

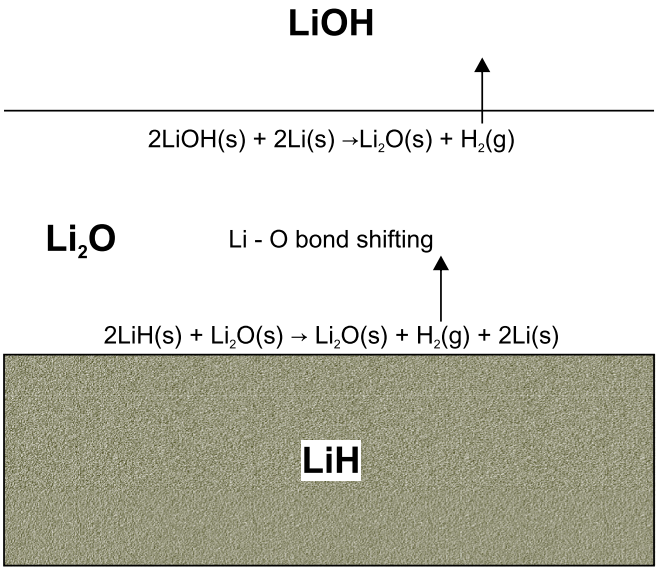


Fig. 9. Schematic of the proposed Li–O bond shift model for the solid state reaction between LiH and LiOH. The chemical equations indicate the reactions occurring at the interface, which sum to the overall solid state reaction. The only gaseous species realised in this scheme is H_2.

The overall reaction is the same as the observed reaction, $LiH + LiOH \rightarrow Li_2O + H_2O$, equation (2b). Diffusion, in this model, is in the sense of bond shifting in the Li_2O layer.

5. Conclusions

Thin films of Li, LiH, Li_2O and LiOH have been synthesised and characterised using AES, XPS and TPD. Characterisation has shown peak shifts in both the Li *KLL* and the O 1s spectra, allowing for identification of these Li compounds. LiH films were then analysed using TPD and characterised using XPS after heating to 675 K; this showed a H_2 desorption peak at 425 K associated with a solid state reaction between adventitious LiH and LiOH, resulting in the formation of Li_2O and H_2. The XPS shows a reduction in the LiOH present and an increase in the amount of Li_2O, providing further evidence of the prevalence of the solid state reaction in very thin corrosion layers. Also present within the TPD are decomposition and sublimation of LiH to Li and H_2 at 825 K.

After exposing LiH films to an excess (450 L) of H_2O, XPS of the O 1s region indicated the presence of LiOH and adsorbed H_2O. The TPD of this film contained two peaks within the H_2O trace, attributed to the desorption of chemisorbed H_2O at 425 K and the decomposition of LiOH into Li_2O and H_2O at 535 K. The former, to our knowledge, is the first observation of desorbing H_2O ad-layers from the LiOH/LiH surface. A fraction of the H_2O liberated from LiOH diffuses through the Li_2O to the LiH surface and reacts with the LiH to form further Li_2O and H_2. The use of multiple ramp rates caused these peaks to shift and shows that the reaction of LiH and H_2O closely follows the decomposition of LiOH at all rates. There was no evidence to suggest that adsorbed H_2O diffuses through LiOH and Li_2O to react with LiH. Reducing the H_2O exposure, such that the XPS of the O 1s region shows no adsorbed H_2O, eliminates the adsorbed H_2O in the TPD profile.

Kissinger analysis was unsuccessful in determining the E_a for the decomposition reactions. The failure was suggested to be a result of the mechanisms involved being of an inappropriate mathematical form. Instead, explicit modelling was carried out to determine E_a and ν for each reaction. The values for E_a were 72.7 kJ mol^{-1} for the desorption of H_2O; 90.1 kJ mol^{-1} for the decomposition of LiOH both with an A3 mechanism; and 81.6 kJ mol^{-1} for the solid state reaction with a D3 mechanism, which are in reasonable agreement with the values obtained for bulk specimens. The increase in E_a for the reaction of LiH with H_2O is reasoned to be an artefact caused by a transport process of H_2O through the Li_2O layer. A model for the solid state reaction is proposed based upon Li–O bond shifting.

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